



7 (a) Show that $\frac{d}{dx} \left(\frac{1}{2}x\sqrt{4-x^2} + 2 \sin^{-1} \left(\frac{1}{2}x \right) \right) = \sqrt{4-x^2}$. [3]

This image shows a blank sheet of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the paper.

(b) Find the solution of the differential equation

$$2\frac{dy}{dx} + \frac{y}{2+x} = 2\sqrt{2-x}$$

for which $y = \frac{1}{2}$ when $x = 1$. Give your answer in an exact form. [8]

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